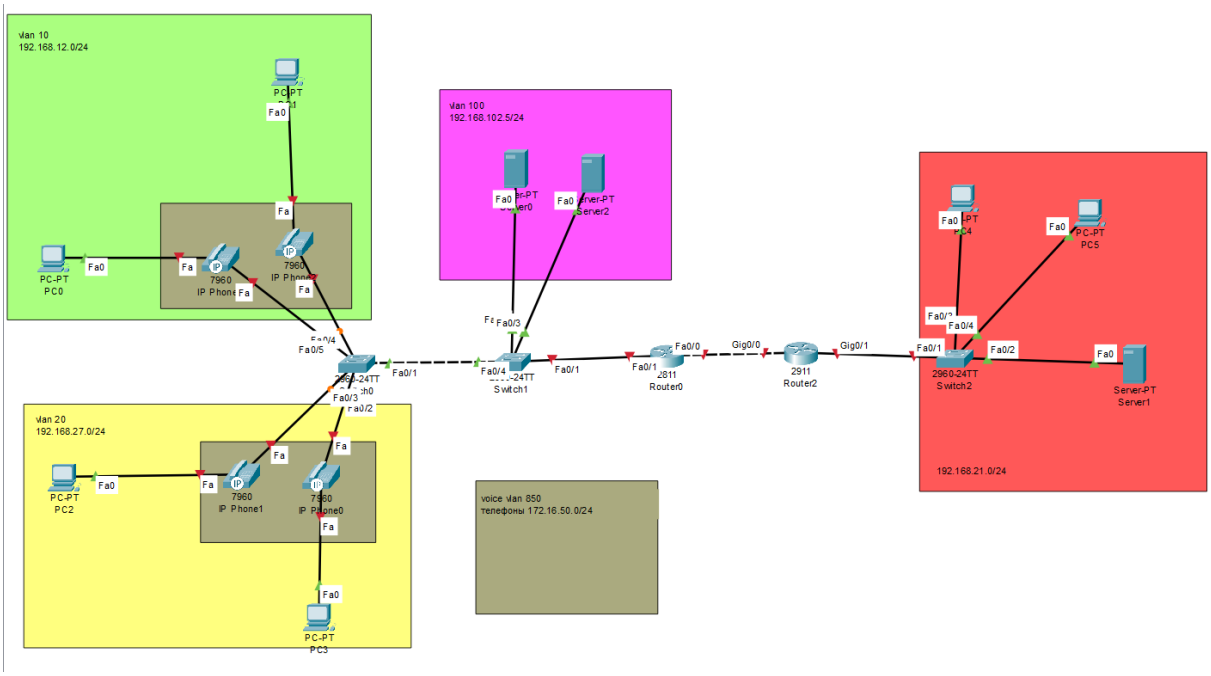
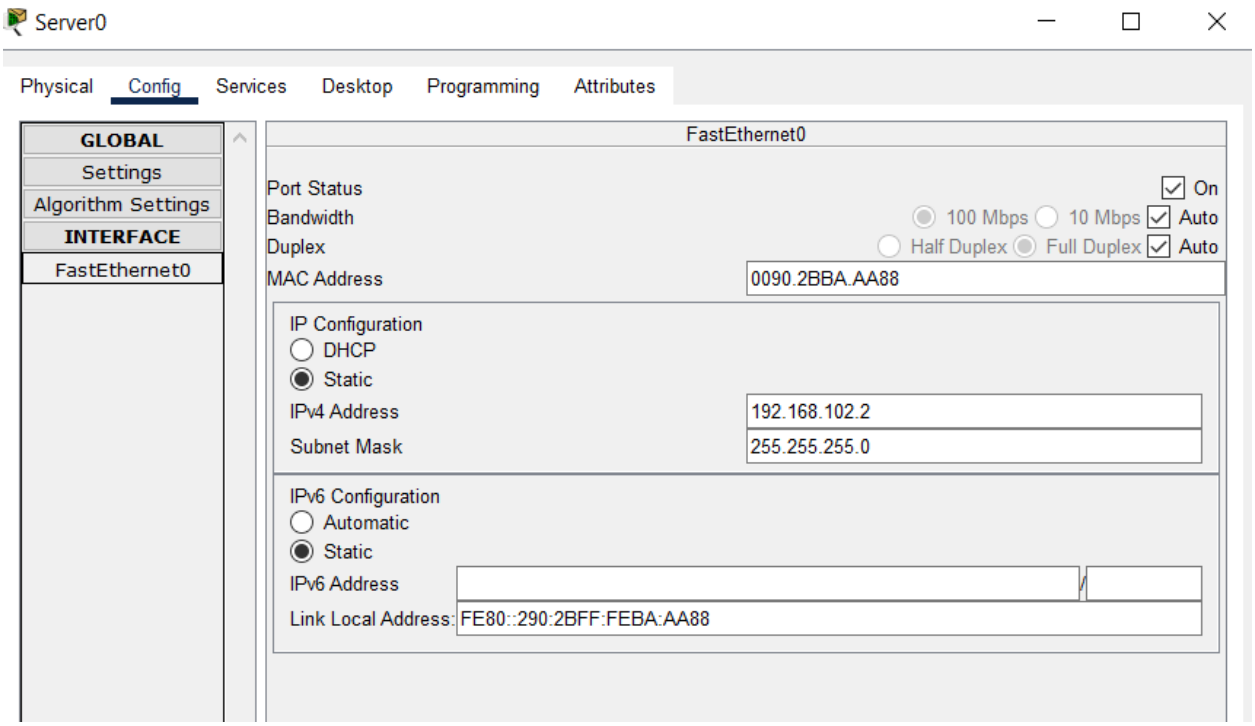
**Практическая работа №4**

Настройка топологии в CPT:

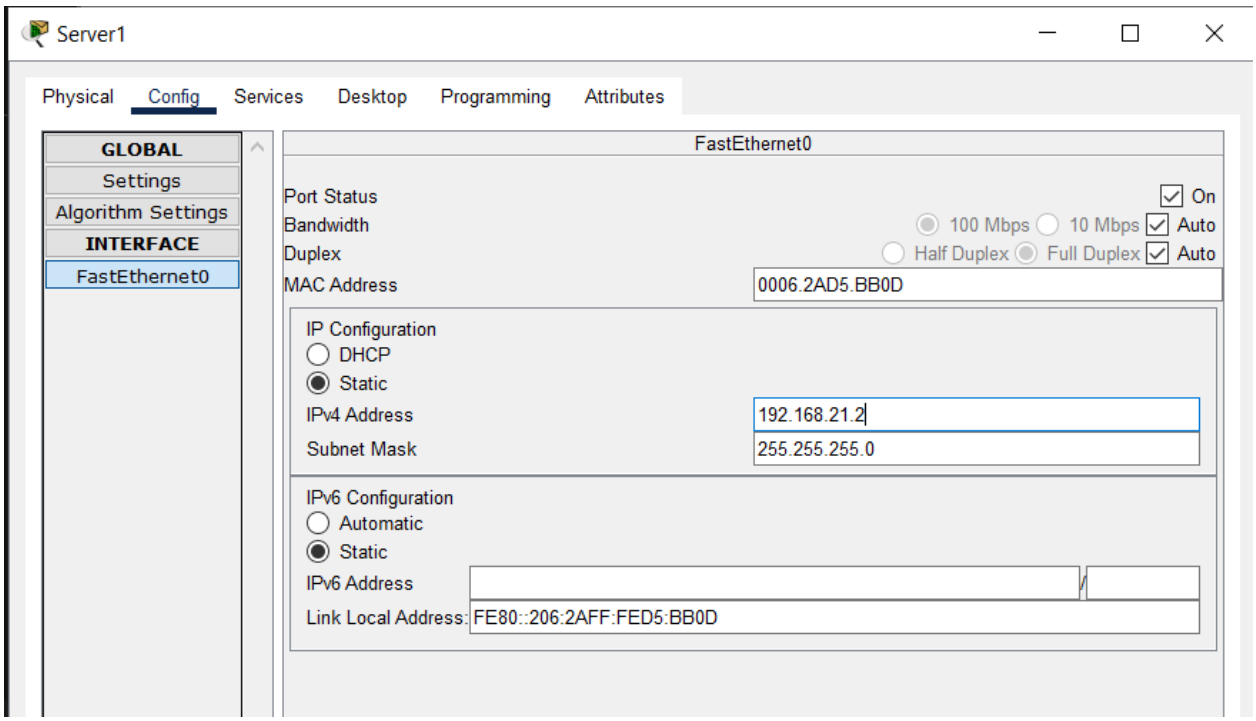



Настройка компютеров, серверов и свитчей:

Server0:




Server1:


Server1
Physical
Config
Services
Desktop
Programming
Attributes
GLOBAL
Settings
Algorithm Settings
INTERFACE
FastEthernet0
FastEthernet0
Port Status
On
Bandwidth
100 Mbps
10 Mbps
Auto
Duplex
Half Duplex
Full Duplex
Auto
MAC Address
0006.2AD5.BB0D
IP Configuration
DHCP
Static
IPv4 Address
192.168.21.2
Subnet Mask
255.255.255.0
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address
FE80::206:2AFF:FED5:BB0D

Switch0:

```
Switch>enable
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#exit
Switch(config)#vlan 20
Switch(config-vlan)#exit
Switch(config)#vlan
% Incomplete command.
Switch(config)#vlan 100
Switch(config-vlan)#exit
Switch(config)#vlan 850
Switch(config-vlan)#exit
Switch(config)#exit
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#interface range fa0/2-5
                ^
% Invalid input detected at '^' marker.

Switch#inte range fa0/2-5
Switch#interface range fa0/2-5
                ^
% Invalid input detected at '^' marker.

Switch#interf
Switch#interface range fa0/2-5
                ^
% Invalid input detected at '^' marker.

Switch#
Switch#inter
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#interface ra
Switch(config)#interface range fa0/2-5
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 10
Switch(config-if-range)#switchport access vlan 20
Switch(config-if-range)#switchport voice vlan 850
Switch(config-if-range)#exit
Switch(config)#intef
Switch(config)#interf
Switch(config)#interface fa0/1
Switch(config-if)#swi
Switch(config-if)#switchport mode trunk

Switch(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
```

Switch1:

```
Switch>enable \
             ^
% Invalid input detected at '^' marker.

Switch>enable
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#valn 10
               ^
% Invalid input detected at '^' marker.

Switch(config)#vlan 10
Switch(config-vlan)#exit
Switch(config)#vlan 20
Switch(config-vlan)#exit
Switch(config)#vlan 100
Switch(config-vlan)#exit
Switch(config)#vlan 850
Switch(config-vlan)#exit
Switch(config)#interface range fa0/2-3
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 100
Switch(config-if-range)#switchport access voice vlan 850
                                              ^
% Invalid input detected at '^' marker.

Switch(config-if-range)#
Switch(config-if-range)#switchport voice vlan 850
Switch(config-if-range)#exit
Switch(config)#interface range fa0/1
Switch(config-if-range)#switchport mode trunk
Switch(config-if-range)#exit
Switch(config)#interface range fa0/4
Switch(config-if-range)#switchport mode trunk
Switch(config-if-range)#exit
Switch(config)#
```

Настройка маршрутизатора (Router0 и 1):

Router0:

IOS Command Line Interface

```
% Configuring IP routing on a LAN subinterface is only allowed if that
subinterface is already configured as part of an IEEE 802.10, IEEE 802.1Q,
or ISL vLAN.

R0(config-subif)#
R0(config-subif)#encapsulation dot1Q 100
R0(config-subif)#ip address 192.168.102.1 255.255.255.0
R0(config-subif)#exit
R0(config)#interface fa0/1.850
R0(config-subif)#
%LINK-5-CHANGED: Interface FastEthernet0/1.850, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1.850, changed state to up

R0(config-subif)#encapsulation dot1Q 850
R0(config-subif)#ip address 172.16.50.1 255.255.255.0
R0(config-subif)#exit
R0(config)#telephony-service
R0(config-telephony)#max-ephones 4
R0(config-telephony)#max-dn 4
R0(config-telephony)#ip source-address 172.16.50.1 port 2000
R0(config-telephony)#auto assign 1 to 4
R0(config-telephony)#exit
R0(config)#ephone-dn 1
R0(config-ephone-dn)#%LINK-3-UPDOWN: Interface ephone_dsp DN 1.1, changed state to up

R0(config-ephone-dn)#number 77261
R0(config-ephone-dn)#exit
R0(config)#ephone-dn 2
R0(config-ephone-dn)#%LINK-3-UPDOWN: Interface ephone_dsp DN 2.1, changed state to up

R0(config-ephone-dn)#number 77262
R0(config-ephone-dn)#exit
R0(config)#ephone-dn 3
R0(config-ephone-dn)#%LINK-3-UPDOWN: Interface ephone_dsp DN 3.1, changed state to up

R0(config-ephone-dn)#number 77263
R0(config-ephone-dn)#exit
R0(config)#ephone-dn 4
R0(config-ephone-dn)#%LINK-3-UPDOWN: Interface ephone_dsp DN 4.1, changed state to up

R0(config-ephone-dn)#number 77264
R0(config-ephone-dn)#exit
R0(config)#interface fa0/0
R0(config-if)#ip add
R0(config-if)#ip address 10.10.10.1 255.255.255.252
R0(config-if)#no s
R0(config-if)#no shu
R0(config-if)#no shutdown

R0(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

R0(config-if)#exit
R0(config)#router ospf 1
R0(config-router)#network 192.168.12.0 0.0.0.255 area 0
R0(config-router)#network 192.168.27.0 0.0.0.255 area 0
R0(config-router)#network 192.168.102.0 0.0.0.255 area 0
R0(config-router)#network 10.10.10.0 0.0.0.3 area 0
R0(config-router)#exit
```

Эхо тестирование:

| Тестирующий Узел (IP) | Тестируемые узлы (IP-адреса) | | | | |
|---|---|---|---|---|---|
| | 192.168.12.4 | 192.168.102.2 | 192.168.12.1 | 192.168.27.1 | 192.168.102.1 |
| 192.168.12.4 | + | + | + | + | + |
| 192.168.12.3 | + | + | + | + | + |
| 192.168.27.4 | + | + | + | + | + |
| 192.168.27.3 | + | + | + | + | + |

| Тестирующий Узел (IP) | Тестируемые узлы (IP-адреса) | |
|---|---|---|
| | 192.168.21.5 | 192.168.21.4 |
| 192.168.12.4 | + | + |
| 192.168.27.4 | + | + |
| 192.168.102.2 | + | + |

Настройка ACL:

Разрешим доступ к управлению роутером только для PC2:

```
R0>enable
Password:
R0#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R0(config)#ip access-list standart Telnet
                              ^
% Invalid input detected at '^' marker.

R0(config)#ip access-list standard Telnet
R0(config-std-nacl)#permit host 192.168.27.4
R0(config-std-nacl)#dany any
                    ^
% Invalid input detected at '^' marker.

R0(config-std-nacl)#deny any
R0(config-std-nacl)#exit
R0(config)#line vty 0 15
R0(config-line)#access-class Telnet in [cite: 241]
                                       ^
% Invalid input detected at '^' marker.

R0(config-line)#access-class Telnet in
```

Расширенный ACL для Web-сервера:

```
R0>enable
Password:
R0#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R0(config)#ip access-list extended From-Web
R0(config-ext-nacl)#permit tcp 192.168.27.0 host 192.168.21.5 www
                                            ^
% Invalid input detected at '^' marker.

R0(config-ext-nacl)#permit tcp 192.168.27.0 host 192.168.21.5 eq www
                                            ^
% Invalid input detected at '^' marker.

R0(config-ext-nacl)#permit tcp 192.168.27.0 0.0.0.255 host 192.168.21.5 eq www
R0(config-ext-nacl)#permit ip host 192.168.27.3 host 192.168.21.5
R0(config-ext-nacl)#deny ip 192.168.27.0 0.0.0.255 host 192.168.21.5
R0(config-ext-nacl)#
```